Daily apple versus dried plum: impact on cardiovascular disease risk factors in postmenopausal women.

BACKGROUND: Evidence suggests that consumption of apple or its bioactive components modulate lipid metabolism and reduce the production of proinflammatory molecules. However, there is a paucity of such research in human beings. OBJECTIVE: Women experience a lower rate of cardiovascular disease before menopause compared with men. However, after the onset of menopause, the risk of cardiovascular disease increases drastically due to ovarian hormone deficiency. Hence, we conducted a 1-year clinical trial to evaluate the effect of dried apple vs dried plum consumption in reducing cardiovascular disease risk factors in postmenopausal women. DESIGN: One-hundred sixty qualified postmenopausal women were recruited from the greater Tallahassee, FL, area during 2007-2009 and were randomly assigned to one of two groups: dried apple (75 g/day) or dried plum (comparative control). Fasting blood samples were collected at baseline, 3, 6, and 12 months to measure various parameters. Physical activity recall and 7-day dietary recall were also obtained. RESULTS: Neither of the dried fruit regimens significantly affected the participants' reported total energy intake throughout the study period. On the contrary, women who consumed dried apple lost 1.5 kg body weight by the end of the study, albeit not significantly different from the dried plum group. In terms of cholesterol, serum total cholesterol levels were significantly lower in the dried apple group compared with the dried plum group only at 6 months. Although dried plum consumption did not significantly reduce serum total cholesterol and low-density lipoprotein cholesterol levels, it lowered their levels numerically by 3.5% and 8%, respectively, at 12 months compared with baseline. This may explain the lack of significance observed between the groups. However, within the group, women who consumed dried apple had significantly lower serum levels of total cholesterol and low-density lipoprotein cholesterol by 9% and 16%, respectively, at 3 months compared with baseline. These serum values were further decreased to 13% and 24%, respectively, after 6 months but stayed constant thereafter. The within-group analysis also reported that daily apple consumption profoundly improved atherogenic risk ratios, whereas there were no significant changes in lipid profile or atherogenic risk ratios as a result of dried plum consumption. Both dried fruits were able to lower serum levels of lipid hydroperoxide and C-reactive protein. However, serum C-reactive protein levels were significantly lower in the dried plum group compared with the dried apple group at 3 months. CONCLUSIONS: There were no significant differences between the dried apple and dried plum groups in altering serum levels of atherogenic cholesterols except total cholesterol at 6 months. However, when within treatment group comparisons are made, consumption of 75 g dried apple (about two medium-sized apples) can significantly lower atherogenic cholesterol levels as early as 3 months. Furthermore, consumption of dried apple and dried plum are beneficial to human health in terms of anti-inflammatory and antioxidative properties.